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NPTEL (https://swayam.gov.in/explorer?ncCode=NPTEL) » Advanced Distributed Systems (course)

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Course outline

About NPTEL ()

How does an NPTEL online course work? ()

Week 1 ()

- Lecture 1 (unit?unit=16&lesson=17)
- Lecture 2 (unit?unit=16&lesson=18)

Week 1 : Assignment 1

Assignment not submitted **Due date: 2025-08-06, 23:59 IST.**

1) In a rumor mongering algorithm, there is a/an ____ relationship between the traffic and the residue. ***1 point***

- exponential
- logarithmic
- linear
- quadratic

2) Which of the following statements is/are correct regarding the rumor mongering protocol? ***1 point***

S1: There is a chance that updates might not reach a node
S2: It has a built-in termination protocol.

- S1 is true and S2 is true
- S1 is true and S2 is false
- S1 is false and S2 is true
- S1 is false and S2 is false

3) In the anti-entropy algorithm, if the number of infected nodes in the system is high, then a_____ based algorithm is preferred. ***1 point***

- push
- pull
- push-pull

Quiz: Week 1 : Assignment 1 (assessment?name=112)

Week 2 ()

Week 3 ()

Transcripts ()

Books ()

- pull-push

4) In an epidemic protocol, a/an ____ node has not received the update. ***1 point***

- infective
- removed
- injured
- susceptible

5) Assume an anti-entropy algorithm that distributes an update across 'n' nodes. Any node can contact any other node. How much time does this take (approximately, asymptotically)? ***1 point***

- O(n)
- O(log(n))
- $O(n^2)$
- $O(n^3)$

6) Consider the following statements. ***1 point***

S1: In Napster, a client can provide fake details such as IP addresses.
S2: In Napster, the main coordination entity handles incoming connections from peers.

- S1 is true and S2 is true
- S1 is true and S2 is false
- S1 is false and S2 is true
- S1 is false and S2 is false

7) In Gnutella, the ____ entity manages connections with other Gnutella peers. ***1 point***

- Connection handler
- Co-ordination instance
- Download instance
- Upload instance

8) The co-ordination instance in Gnutella sends the ____ message back to the original client after updating the local database with information about remote peers. ***1 point***

- ping
- pong
- search
- client push

9) In Napster, a client connects to a/the ____ broker. ***1 point***

- Least busy
- Geographically nearest

- Popular
- Random

10) 10. Which of the following statements is/are incorrect? ***1 point***

- Gnutella is resilient to network partitions.
- Napster ensures load balancing by finding the best broker.
- Gnutella is scalable due to its distributed architecture.
- None of the options

You may submit any number of times before the due date. The final submission will be considered for grading.

Submit Answers